

Securitize Solana Vault Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

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May 23, 2025

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Securitize on Securitize Solana Vault. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated May 23, 2025, was delivered to Securitize, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 6 days, the Cyfrin team conducted an audit on the Securitize Solana Vault code provided by Securitize. The findings consist of 1 High, 1 Medium & 5 Low severity issues with the remainder being informational.

Summary

Project Name	Securitize Solana Vault
Repository	bc-solana-vault-sc
Commit	e7244a875189...
Audit Timeline	May 12th - May 19th, 2025
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	1	1	0
Medium Risk	1	0	1
Low Risk	5	5	0
Informational	3	3	0
Gas Optimizations	0	0	0
Total	10	9	1

6 Audit Scope

Cyfrin conducted an audit of the Securitize Solana Vault based on the code present in the repository commit hash e7244a8. The following files construct the scope of the audit:

```
programs/nav-provider-interface/src/lib.rs
programs/redemption-interface/src/lib.rs
programs/sc-vault/src/constants.rs
programs/sc-vault/src/errors.rs
programs/sc-vault/src/events.rs
programs/sc-vault/src/instructions/admin/add_liquidator.rs
programs/sc-vault/src/instructions/admin/add_operator.rs
programs/sc-vault/src/instructions/admin/add_share_metadata.rs
programs/sc-vault/src/instructions/admin/change_admin.rs
programs/sc-vault/src/instructions/admin/initalize.rs
programs/sc-vault/src/instructions/admin/mod.rs
programs/sc-vault/src/instructions/admin/pause.rs
programs/sc-vault/src/instructions/admin/revoke_liquidator.rs
programs/sc-vault/src/instructions/admin/revoke_operator.rs
programs/sc-vault/src/instructions/admin/set_liquidation_open_to_public.rs
programs/sc-vault/src/instructions/admin/unpause.rs
programs/sc-vault/src/instructions/admin/update_nav_provider.rs
programs/sc-vault/src/instructions/liquidator/liquidate.rs
programs/sc-vault/src/instructions/liquidator/mod.rs
programs/sc-vault/src/instructions/mod.rs
```

```
programs/sc-vault/src/instructions/operator/accounts.rs
programs/sc-vault/src/instructions/operator/deposit.rs
programs/sc-vault/src/instructions/operator/mod.rs
programs/sc-vault/src/instructions/operator/redeem.rs
programs/sc-vault/src/instructions/user/accounts.rs
programs/sc-vault/src/instructions/user/convert_to_assets.rs
programs/sc-vault/src/instructions/user/convert_to_shares.rs
programs/sc-vault/src/instructions/user/get_share_value.rs
programs/sc-vault/src/instructions/user/is_admin.rs
programs/sc-vault/src/instructions/user/is_liquidator.rs
programs/sc-vault/src/instructions/user/is_operator.rs
programs/sc-vault/src/instructions/user/mod.rs
programs/sc-vault/src/lib.rs
programs/sc-vault/src/states/mod.rs
programs/sc-vault/src/states/vault_counter.rs
programs/sc-vault/src/states/vault_state.rs
programs/sc-vault/src/utils/conversions.rs
programs/sc-vault/src/utils/get_rate.rs
programs/sc-vault/src/utils/get_vault_authority_signer.rs
programs/sc-vault/src/utils/math.rs
programs/sc-vault/src/utils/mod.rs
programs/sc-vault/src/utils/transfer_from_vault.rs
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

The Securitize Vault is a Solana-based implementation of a tokenized vault system inspired by the ERC4626 standard. It enables users to deposit DS Tokens (regulated digital securities) and receive Vault Tokens in return, which are SPL tokens representing a share of the underlying assets. These Vault Tokens can later be redeemed for the original DS Tokens or, alternatively, liquidated into stablecoins via external liquidity providers.

The architecture is modular and extensible, implemented using the Anchor framework and organized into multiple programs. The primary `sc-vault` program contains the core logic for asset management, while additional interface programs define interactions with external NAV providers and redemption systems. Mock programs for both NAV and redemption are included to facilitate testing and simulation.

Key components include a persistent `VaultState` account that holds the vault's configuration and tracks links to the NAV and redemption providers. The NAV provider supplies the asset valuations used in share conversions, and the redemption interface allows for flexible strategies in converting Vault Tokens back into DS Tokens.

The system is built with compatibility for both Token Program v1 and Token-2022, supporting token extensions such as transfer hooks for KYC enforcement. Instructions cover the full lifecycle of the vault, from initialization and configuration to asset deposit, share redemption and liquidation.

9 Attestation

Cyfrin attests that it performed a security review of the Securitize Solana Vault code provided by Securitize.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-05-23-cyfrin-securitize-solana-vault-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Securitize has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=20b0c77bb1834baf1525efce63f0f38490c4534bd1e931452d3769b45e9e104f
F=2025-05-23-cyfrin-securitize-solana-vault-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.